

Ruihan Henry Liu

PH.D. CANDIDATE · UNIVERSITY OF CALIFORNIA, BERKELEY

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Research Interests

I work at the intersection of observational and computational cosmology. My research focuses on the large-scale structure of the Universe and the cosmic microwave background, with an emphasis on probing baryonic matter and feedback through the Sunyaev-Zel'dovich effects and weak gravitational lensing. I make use of data from cosmological surveys such as **ACT, DESI, and HSC**, together with hydrodynamical simulations including **IllustrisTNG, SIMBA, and CAMELS**. I also develop and apply Bayesian inference and machine learning methods for cosmological data analysis.

Education

University of California, Berkeley

DOCTOR OF PHILOSOPHY (PH.D.) IN PHYSICS

Advisors: Uroš Seljak and Simone Ferraro

MASTER OF ARTS (MA) IN PHYSICS

Berkeley, California

Expected May 2027

Dec. 2024

University of British Columbia

B.Sc. (HONOURS) IN PHYSICS AND MATHEMATICS

Thesis: *On Bayesian Baseline Removal and Signal Extraction Methods in Astronomy*

Advisor: Douglas Scott

Vancouver, B.C., Canada

May 2021

Research Experience

University of California, Berkeley

GRADUATE RESEARCHER

- Cosmology: Large Scale Structure and the Cosmic Microwave Background. Focusing on probes of baryonic matter through the Sunyaev-Zel'dovich effects and weak lensing, as well as computational and statistical approaches to cosmology.

Berkeley, California

Sept. 2021 - Present

University of California, Berkeley · VISITING STUDENT RESEARCHER (ADVISOR: U. SELJAK)

May 2020 - Aug. 2020

Canadian Institute for Theoretical Astrophysics · UNDERGRADUATE RESEARCHER (ADVISOR: P. C. BREYSSE)

May 2019 - Aug. 2019

University of British Columbia · NSERC USRA UNDERGRADUATE RESEARCHER (ADVISOR: D. SCOTT)

May 2018 - Jul. 2019

Imperial College London · IROP EXCHANGE RESEARCH STUDENT (ADVISOR: R. TROTTE)

Jul. 2018 - Aug. 2018

NRC Herzberg Institute of Astrophysics · CO-OP MACHINE LEARNING STUDENT (ADVISORS: S. FABBRO AND A.

McCONNACHIE)

Jan. 2018 - Apr. 2018

Publications and Preprints

First Author Publications:

- [1] Liu, R. H., Ferraro, S., Hadzhiyska, B., **Baryons Beyond Halos: Unbound Gas, Stars, and the Sunyaev-Zel'dovich Effect, in preparation, to be submitted to Phys. Rev. D.**
- [2] Liu, R. H., Seljak, U., Ferraro, S., Hadzhiyska, B., *et al.*, **Model-Independent Measurement of Baryon Gas Fractions through Galaxy-Galaxy Lensing and the Kinematic Sunyaev-Zel'dovich Effect, Submitted to Phys. Rev. D.**, arXiv:2609.22466.
- [3] Liu, R. H., Hadzhiyska, B., Ferraro, S., Bose, S., Hernández-Aguayo, C., **Fast Baryonic Field Painting for Sunyaev-Zel'dovich Analyses: Transfer Function vs. Hybrid Effective Field Theory, Phys. Rev. D 113, 063558 (2026)**, arXiv:2504.11794.
- [4] Liu, R. H., Ferraro, S., Schaan, E., *et al.*, **Measurements of the Thermal Sunyaev-Zel'dovich Effect with ACT and DESI Luminous Red Galaxies, Phys. Rev. D 112, 083561 (2025)**, arXiv:2502.08850.
- [5] Liu, R. H. and Breysse, P. C., **Coupling parsec and gigaparsec scales: primordial non-Gaussianity with multi-tracer intensity mapping, Phys. Rev. D 103, 063520 (2021)**, arXiv:2002.10483.

- [6] **Liu, R. H.**, Hill, R., Scott, D., Almaini, O., An, F., Gubbels, C., Hsu, L., Lin, L., Smail, I., Stach, S., **A machine learning approach for identifying the counterparts of submillimetre galaxies and applications to the GOODS-North field**, *MNRAS* **489**, 1770 (2019), arXiv:1901.09594.

Other Publications:

- [7] Gong, Y., Gallardo, P. A., Bean, R., Moore, J., Vavagiakis, E. M., ..., **Liu, R. H.**, ..., *et al.*, **Detection of the pairwise kinematic Sunyaev-Zel’dovich effect and pairwise velocity with DESI DR1 galaxies and ACT DR6 and Planck CMB data**, *Phys. Rev. D* **113**, 063538 (2026), arXiv:2511.23417.
- [8] Hadzhiyska, B., Ferraro, S., Ried Guachalla, B., Schaan, E., ..., **Liu, R. H.**, ..., *et al.*, **Evidence for large baryonic feedback at low and intermediate redshifts from kinematic Sunyaev-Zel’dovich observations with ACT and DESI photometric galaxies**, *Phys. Rev. D* **112**, 083509 (2025), arXiv:2407.07152.

Talks and Presentations

Seminars:

Observational Cosmology Seminar, California Institute of Technology	<i>Pasadena, California</i>
PROBING BARYON FEEDBACK WITH THE SUNYAEV-ZEL'DOVICH EFFECTS AND WEAK LENSING	<i>Sep. 2026</i>
Cosmo Coffee, University of British Columbia	<i>Vancouver, Canada</i>
MODEL-INDEPENDENT MEASUREMENTS OF BARYON FEEDBACK WITH KSZ AND GALAXY-GALAXY LENSING	<i>Sep. 2026</i>
CMB/LSS Seminar, Institute of Astronomy, Cambridge University	<i>Cambridge, United Kingdom</i>
MODEL-INDEPENDENT MEASUREMENTS OF BARYON FEEDBACK WITH KSZ AND GALAXY-GALAXY LENSING	<i>Jul. 2026</i>
CMB Lunch, Lawrence Berkeley National Laboratory	<i>Berkeley, California</i>
MODEL-INDEPENDENT MEASUREMENTS OF BARYON FEEDBACK WITH KSZ AND LENSING	<i>Apr. 2026</i>
Stanford Cosmology Seminar	<i>Stanford, California</i>
PROBING BARYONIC MATTER IN THE UNIVERSE WITH THE SUNYAEV-ZEL'DOVICH EFFECT	<i>Mar. 2025</i>
CMB Lunch, Lawrence Berkeley National Laboratory	<i>Berkeley, California</i>
PROBING BARYONIC MATTER IN THE UNIVERSE WITH THE SUNYAEV-ZEL'DOVICH EFFECT	<i>Jan. 2025</i>
Cosmology Seminar, National Astronomical Observatory of China	<i>Beijing, China</i>
PROBING BARYONIC MATTER IN THE UNIVERSE WITH THE SUNYAEV-ZEL'DOVICH EFFECT	<i>Aug. 2024</i>

Conference and Collaboration Talks:

New Frontiers in Cosmology	<i>A Coruña, Spain</i>
MODEL-INDEPENDENT MEASUREMENTS OF BARYON FEEDBACK WITH KSZ AND GALAXY-GALAXY LENSING	<i>Aug. 2026</i>
DESI Collaboration Summer Meeting	<i>Durham, United Kingdom</i>
MODEL-INDEPENDENT MEASUREMENTS OF BARYON FEEDBACK WITH DESI KSZ AND LENSING	<i>Jul. 2026</i>
From Pen and Paper to Petaflops	<i>San Francisco, California</i>
MODEL-INDEPENDENT MEASUREMENTS OF BARYON FEEDBACK WITH KSZ AND LENSING	<i>May 2026</i>
New Synergies in Multi-Probe Cosmology (Poster)	<i>KITP, Santa Barbara, California</i>
PROBING BOUND AND UNBOUND GAS WITHIN SIMULATIONS TO UNDERSTAND THE SUNYAEV-ZEL'DOVICH EFFECTS	<i>Feb. 2026</i>
CAMELS Collaboration Telecon	<i>Virtual</i>
FAST RECONSTRUCTION OF BARYONIC SIMULATION FIELDS USING CAMELS AND MILLENNIUMTNG	<i>Jan. 2026</i>
DESI Collaboration Winter Meeting	<i>Tucson, Arizona</i>
PROBING BOUND AND UNBOUND GAS WITH THE SUNYAEV-ZEL'DOVICH EFFECTS	<i>Dec. 2025</i>
Learning the Universe Collaboration Telecon	<i>Virtual</i>
COMPARISON OF TRANSFER FUNCTION AND HEFT METHODS FOR BARYON PASTING ON SIMULATIONS	<i>Jan. 2025</i>
ACT Collaboration Telecon	<i>Virtual</i>
THERMAL SUNYAEV ZEL'DOVICH EFFECT WITH ACT AND DESI	<i>Sep. 2024</i>
Cosmology in the Adriatic	<i>Split, Croatia</i>
COMPARISON OF TRANSFER FUNCTION AND HEFT METHODS FOR BARYON PASTING ON SIMULATION	<i>Jul. 2024</i>

DESI Collaboration Summer Meeting

DESI x ACT: CROSS CORRELATION FOR DETECTION OF THE THERMAL SUNYAEV-ZEL'DOVICH EFFECT

Marseille, France

Jul. 2024

DESI Collaboration Winter Meeting

DESI x ACT: CROSS CORRELATION FOR DETECTION OF THE THERMAL SUNYAEV-ZEL'DOVICH EFFECT

Waikoloa, Hawaii

Dec. 2023

Collaboration Memberships

Dark Energy Spectroscopic Instrument (DESI), Collaboration Member

Awards and Honours

Hearts to Humanity Eternal Graduate Research Grant, UC Berkeley	2026
Outstanding Graduate Student Instructor Award, UC Berkeley	2025
NSERC Postgraduate Scholarship – Doctoral (PGS-D), Natural Sciences and Engineering Research Council of Canada	2022 - 2025
Sushant Singh Rajput Memorial Fellowship, UC Berkeley	2021
NSERC Undergraduate Student Research Award, Natural Sciences and Engineering Research Council of Canada	2018, 2019
Trek Excellence Scholarship, UBC	2017 - 2020
Awarded to students in the top 5% of their faculty and year. Received during every eligible term.	
Various Other Undergraduate Scholarships, UBC	2016 - 2021

Teaching and Mentoring

Teaching:

Department of Physics, UC Berkeley

Berkeley, California

GRADUATE STUDENT INSTRUCTOR

2021 - Present

- Taught courses on Electromagnetism and Optics, Bayesian Statistics and Machine Learning, and Introductory Physics.
- Received the Outstanding Graduate Student Instructor Award for teaching.

Department of Physics and Astronomy, University of British Columbia

Vancouver, British Columbia

UNDERGRADUATE TEACHING ASSISTANT

Sept. 2018 - Apr. 2019

- PHYS 101: Energy and Waves (taught for two terms).
- Received the Undergraduate Teaching Excellence Award from the UBC Science Undergraduate Society.

Mentorship:

Linda Jin, Post-baccalaureate Researcher, UC Berkeley (now Ph.D. student, University of Washington)

2025 - Present

Machine learning emulators of baryonic suppression of the matter power spectrum using CAMELS simulations; manuscript in preparation.

Volunteering, Service and Outreach

Astronomy On Tap East Bay

Oakland, California

SPEAKER

2026

UC Berkeley Astro Night

Berkeley, California

VOLUNTEER

2024 - 2025

UofT AstroTours

Toronto, Ontario

VOLUNTEER

Jun. 2019 - Aug. 2019

Skills

Programming Computing Languages

Python (NumPy, SciPy, JAX, PyTorch, scikit-learn), C++, R, MATLAB, Mathematica
High-performance computing (NERSC), MPI, git, Linux
English (Native Proficiency), Mandarin Chinese (Limited Working Proficiency)